

Finite Communication and the Nonmonotone Value of Mixing

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Abstract

This paper studies social learning when agents hold real-valued private assessments but communicate through finite messages. Because messages change only when assessments cross category thresholds, private evidence can affect an agent's assessment without becoming visible to others. We show that this finite-communication friction can make social mixing nonmonotone. More mixing helps lagging agents hear from agents who are closer to the true state, but it also pulls those state-aligned agents toward the population average before their assessments are fully reflected in messages. In a block-mixing society with two initially different groups, low mixing preserves segmentation, moderate mixing generates state-aligned consensus, and excessive mixing generates non state-aligned consensus. The last outcome is inefficient because the society becomes trapped in an intermediate message even though private evidence continues to arrive. Thus, under finite communication, better connectivity need not improve learning: too much averaging can suppress the disagreement needed for informative message changes to occur.

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1 Introduction

1.1 Overview

Many economic and social environments do not transmit assessments at the same resolution at which those assessments are formed. Consider a group of firms that monitor a common supplier. Each procurement manager continues to learn from her own transactions: deliveries may be late, defects may appear, refunds may increase, or product quality may improve. These observations change the manager’s internal assessment of the supplier. But firms may not communicate that assessment as a precise number. Instead, they may use a small set of standardized messages, such as “Reliable,” “Watchlist,” and “High Risk.” A manager’s assessment can move substantially inside one category without changing the message that other firms observe. Unless her assessment crosses a threshold, the public message remains the same.

This distinction matters because social learning depends on what becomes visible to others. If firms combine their own experience with the messages sent by other firms, then new private evidence affects the rest of the network only when it changes a message. Evidence that moves an internal assessment within the same category remains socially invisible. The supplier’s perceived quality can therefore become stale, not because firms stop receiving information, but because the information they receive is not transmitted through the finite message system.

The same friction appears in many settings. Credit ratings, risk buckets, grades, platform labels, medical triage categories, and internal performance classifications all convert richer assessments into finite messages. These systems are useful because they make communication simple and comparable. But they also create thresholds. A change in the underlying assessment matters for social learning only if it is large enough to change the message. This paper studies how that finite-communication friction interacts with social mixing.

In standard models where agents communicate real-valued beliefs or actions, stronger mixing is usually a force for learning. Information spreads more widely, local noise is averaged out, and disagreement is reduced more quickly (Golub and Jackson, 2010; Golub and Jackson, 2012). Under finite communication, the same logic is incomplete. Mixing still helps agents hear from others, but it also compresses their latent assessments. This compression can be harmful because message changes require threshold crossings. If mixing pulls agents toward a common intermediate assessment before enough agents cross into the state-aligned message, the society may become locked into an uninformative common message.

The central mechanism is therefore simple. Agents privately hold real-valued latent scores,

but they publicly send finite messages. Scores can move every period as agents receive private evidence, while messages change only when scores cross cutpoints. Network mixing averages decoded messages, not latent scores. As a result, faster mixing can make agents' social inputs more similar even when their private information is still evolving. Once a large group enters the same communication bin, and that bin is robust to private shocks and outside exposure, the group continues to send the same message. The trap is not caused by the absence of private evidence. It is caused by the failure of private evidence to become socially visible.

The model formalizes this idea in a repeated-learning environment with a fixed binary state. Each agent has a latent score, interpreted as a real-valued assessment of the state. In each period, the agent observes finite messages from others, aggregates the decoded values through a social exposure matrix, receives fresh private evidence, and updates her score. The key friction is that agents do not transmit scores directly. A common codebook maps each score into one of finitely many messages. Hence the social part of the update depends on the messages induced by others' scores, rather than on the scores themselves.

The first result isolates the role of finite communication. If agents instead communicate their latent scores directly, then the same updating rule leads to asymptotic learning in every fixed irreducible society. Private evidence accumulates in scores, and social averaging spreads that accumulated evidence through the network. Thus the learning failure studied in the paper does not come from the updating rule alone. It comes from replacing real-valued communication with finite messages.

The second result gives a general same-message trapping criterion. Faster mixing reduces cross-sectional dispersion in latent scores. If the population mean lies safely inside a communication bin, low dispersion implies that a large fraction of agents send the same message. If this large same-message set is sufficiently cohesive, and if the bin is robust to the largest possible private shock and outside-message exposure, then the set remains in that bin forever. When the bin is not a state-aligned bin, this creates a persistent error, because agents trapped in that bin never reach an extreme score that would correspond to learning the true state.

The third result studies which message is selected. We consider a block-mixing society with two initially different groups. One group starts with the state-aligned message, while the other starts with a lower message. The mixing parameter controls cross-group exposure. When mixing is low, the groups remain segmented: the initially state-aligned group stays high, and the lagging group stays low. When mixing is moderate, the state-aligned group pulls the lagging group upward, and the society reaches state-aligned consensus. When mixing is excessive, however, both groups are pulled into the interior bin before high-message

consensus is reached. Interior consensus then becomes absorbing.

This delivers the main implication of the paper: the value of social mixing is nonmonotone under finite communication. More mixing is beneficial at first because it exposes lagging agents to better-informed agents. But too much mixing can suppress the disagreement needed for informative message changes to occur. In the block environment, moderate mixing generates lower error than both low-mixing segmentation and high-mixing interior consensus. Better connectivity therefore need not improve learning when communication is finite.

The paper’s contribution is not to show that social mixing is bad. Rather, it identifies a specific transmission channel through which mixing can backfire. With real-valued communication, disagreement can be averaged and transmitted at the same time. With finite communication, these two roles come apart. Averaging reduces disagreement, but transmission requires enough disagreement to push some agents across message cutpoints. When averaging is too strong relative to the resolution of the message system, society may converge to a stable but insufficiently informative message.

1.2 Related literature

This paper contributes to the literature on social learning in networks. The benchmark tradition begins with [DeGroot \(1974\)](#), where agents repeatedly average the opinions of their neighbors. Subsequent work studies when such averaging aggregates dispersed information and when it fails. [DeMarzo et al. \(2003\)](#) show how persuasion bias can lead agents to overweight repeated information, while [Golub and Jackson \(2010\)](#) characterize wisdom-of-crowds properties in large DeGroot societies. [Golub and Jackson \(2012\)](#) show how homophily affects the speed of learning and convergence, and [Chandrasekhar et al. \(2020\)](#) provide experimental evidence on network learning models. This paper keeps the repeated averaging structure but changes what is averaged. Agents do not average real-valued assessments directly; they average decoded finite messages. This distinction is central. With real-valued communication, mixing spreads accumulated evidence. With finite communication, mixing also compresses latent disagreement before disagreement necessarily becomes visible through message changes.

The paper is also related to Bayesian and observational models of social learning. In canonical herding and cascade models, agents observe predecessors’ actions and may eventually stop learning from private signals because public behavior becomes insufficiently informative ([Banerjee, 1992](#); [Bikhchandani et al., 1992](#); [Smith and Sørensen, 2000](#)). Network versions study how Bayesian agents learn from observations generated by a social structure ([Acemoglu et al., 2011](#)). The mechanism here is different. Agents continue to receive informative

private evidence, and the failure is not driven by strategic inference from actions. The failure comes from finite communication: private evidence may move latent scores without changing messages, so it does not become socially visible. The resulting trap is a transmission failure rather than a stopping of private information acquisition.

A closer connection is with work on coarse, categorical, or otherwise limited communication. [Lizzeri \(1999\)](#) studies information revelation by certification intermediaries, where disclosure policies can be coarse. [Onuchic and Ray \(2023\)](#) analyze how categories convey value. [Xu \(2025\)](#) studies social learning when agents observe coarse signals of others’ actions. [Amir et al. \(2025\)](#) introduce granular DeGroot dynamics, where opinions are rounded and the resulting dynamics can differ from standard DeGroot updating. This paper shares the focus on limited message spaces, but the object of interest is the interaction between finite communication and network mixing. Agents privately maintain real-valued latent scores, while others observe only finite messages generated by those scores. This separation between latent scores and transmitted messages produces same-message traps and makes the value of mixing nonmonotone.

The block-mixing analysis relates to work on homophily and group structure in networks. Stochastic block models provide a standard way to represent group-based interaction patterns ([Holland et al., 1983](#)). In economics, [Currarini et al. \(2009\)](#) model friendship formation with homophily, and [Golub and Jackson \(2012\)](#) show how homophily affects learning speed. The block environment in this paper uses the same broad idea of group-based exposure, but for a different purpose. The block structure isolates a single mixing parameter that changes cross-group exposure. This makes transparent the paper’s main force: low mixing preserves segmentation, moderate mixing generates state-aligned consensus, and excessive mixing generates interior consensus.

Finally, the paper is related to work on learning failures and traps. [Liang and Mu \(2020\)](#) study learning traps that arise when information sources are complementary. [Esponda and Pouzo \(2016\)](#) develop Berk–Nash equilibrium as a framework for learning under misspecified models. [Frick et al. \(2020\)](#) show how misinterpreting others can make social learning fragile, and [Frick and Ishii \(2024\)](#) study innovation adoption by forward-looking social learners. The trap in this paper is distinct from these mechanisms. Agents need not be misspecified, strategic, or forward-looking, and private evidence continues to arrive. The problem is that finite messages create cutpoints, and social exposure aggregates decoded messages rather than latent scores. When mixing is too strong relative to the resolution of the message system, averaging can suppress the message changes through which private evidence would otherwise become socially transmitted.

The rest of the paper proceeds as follows. Section 2 introduces the model. Section 3 studies the continuous-communication benchmark. Section 4 develops the general same-message trapping mechanism. Section 5 analyzes the block-mixing environment and derives the nonmonotone value of mixing. Section 6 concludes the paper. The appendices A and B contain proofs and additional robustness details.

2 Model

Agents in the model have two objects that should be kept separate. The first is a latent score, a real number that summarizes the agent’s current assessment of the state. The second is a finite message generated from that score. The paper’s friction is that the network observes the message, not the latent score. Thus a score can move inside a communication bin without changing what other agents hear.

This section introduces the environment in the order in which these objects operate. We first define the state, agents, and latent scores. We then define the finite codebook and the network aggregate of decoded values. After that, we introduce private evidence, the score-updating rule, timing, and the accuracy criterion.

2.1 State, agents, and latent scores

Time is discrete, $t = 0, 1, 2, \dots$. There is a binary state $\theta \in \{0, 1\}$, drawn at date 0 with $\Pr(\theta = 1) = \mu_0 \in (0, 1)$ and fixed thereafter. The probability μ_0 describes the ex ante distribution of the state. For each society size $n \geq 2$, the set of agents is $N^{(n)} = \{1, \dots, n\}$. When n is fixed, we write N and suppress the superscript (n) .

Agent i ’s private state variable at date t is a real-valued latent score $\ell_{i,t} \in \mathbb{R}$. Higher values indicate a stronger assessment that the state is $\theta = 1$, while lower values indicate a stronger assessment that the state is $\theta = 0$, which will be explained in the concluding section of this section. The score is kept as a real number even though communication is finite: agents privately carry latent scores, but they publicly transmit only finite messages generated from those scores.

The initial score profile $\ell_0 = (\ell_{1,0}, \dots, \ell_{n,0})$ is deterministic and bounded. More generally, we allow heterogeneous initial scores.¹ These scores are fixed starting conditions for the communication process.

¹This can be interpreted that each agent has their own initial prior instead of the common prior.

2.2 Finite communication and social exposure

Agents do not transmit their latent scores directly. Instead, finite communication is described by a common codebook. The codebook consists of a finite message set, an encoder, and a decoder. The encoder maps each score into one of finitely many messages, and the decoder assigns a score-scale value to each message. This captures environments in which communication is standardized, such as ratings, grades, risk tiers, classifications, or platform labels. The same message has the same decoded meaning for all receivers, and agents do not strategically choose messages to influence others.

Fix $K \geq 2$ and let $\mathcal{M} = \{1, \dots, K\}$ be the finite message set. An encoder is a measurable map $Q : \mathbb{R} \rightarrow \mathcal{M}$, and a decoder is a map $\Psi : \mathcal{M} \rightarrow \mathbb{R}$. The codebook $(\mathcal{M}, Q, \Psi)$ is common across agents and fixed across society sizes. If agent j has score $\ell_{j,t}$ at the communication stage of date t , she sends the message

$$m_{j,t} = Q(\ell_{j,t}),$$

and receivers assign the decoded value

$$y_{j,t} = \Psi(m_{j,t}).$$

Equivalently, the decoded value is $y_{j,t} = \psi(\ell_{j,t})$, where $\psi = \Psi \circ Q$.

The main text focuses on threshold codebooks. Let

$$-\infty = b_0 < b_1 < \dots < b_{K-1} < b_K = +\infty,$$

and define $B_k = (b_{k-1}, b_k]$ for $k = 1, \dots, K$. The encoder assigns message k whenever $\ell \in B_k$, and the decoder assigns the score-scale reconstruction $r_k = \Psi(k)$. Thus

$$\psi(\ell) = r_k \quad \text{for every } \ell \in (b_{k-1}, b_k], \quad k = 1, \dots, K.$$

The decoded values are bounded: $r_k \in [\underline{r}, \bar{r}]$ for every k . We also impose bin consistency:

$$r_1 \leq b_1, \quad r_k \in [b_{k-1}, b_k] \quad \text{for } k = 2, \dots, K-1, \quad r_K \geq b_{K-1}.$$

Bin consistency says that the decoded value assigned to a message lies in the score region represented by that message. It also implies that decoded values are weakly increasing: $r_1 \leq r_2 \leq \dots \leq r_K$.

Because ψ is step-valued, within-bin movements in scores are socially invisible. A message

changes only when a score crosses a cutpoint. We say that agent j has a cutpoint crossing between dates $t - 1$ and t if $Q(\ell_{j,t}) \neq Q(\ell_{j,t-1})$. If no agent has a cutpoint crossing, then all messages and decoded values remain unchanged even if latent scores have moved.

Social exposure is represented by an $n \times n$ matrix $P^{(n)} = (P_{ij}^{(n)})_{i,j \in N^{(n)}}$. For each n , $P^{(n)}$ is row-stochastic and has no self-weights:

$$P_{ij}^{(n)} \geq 0, \quad P_{ii}^{(n)} = 0, \quad \sum_{j=1}^n P_{ij}^{(n)} = 1 \quad \text{for every } i.$$

The coefficient $P_{ij}^{(n)}$ is the weight that agent i assigns to agent j 's decoded value. The restriction $P_{ii}^{(n)} = 0$ reflects that each agent directly carries over her own latent score in the updating rule below, while finite communication applies to information received from others.

At date t , agent i 's social aggregate is

$$S_{i,t} = \sum_{j=1}^n P_{ij} \psi(\ell_{j,t}).$$

Thus the network aggregates decoded values rather than latent scores. This distinction is central. Network mixing can average what agents communicate even when their latent scores contain within-bin variation that is invisible to others.

Remark 1. Finite communication separates latent scores from social transmission. Two score profiles can generate the same messages whenever each agent's score remains in the same communication bin. In that case, the network observes the same decoded values and forms the same social aggregates, even though the underlying latent scores may differ.²

2.3 Private evidence and score dynamics

After date- t communication and before the next score is formed, agent i receives private evidence. We summarize the informational content of that evidence by a private increment $\Delta_{i,t+1}$. For a dominated signal structure with conditional densities $f_i(\cdot \mid \theta)$, this increment can be written as the log-likelihood ratio

$$\Delta_{i,t+1} = \log \frac{f_i(x_{i,t+1} \mid \theta = 1)}{f_i(x_{i,t+1} \mid \theta = 0)}.$$

²Suppose $\ell_i \neq \ell_j$ but $\ell_i, \ell_j \in (b_{k-1}, b_k)$. Then, the network will observe the same encoded message and give the same decoded values.

Positive increments favor $\theta = 1$, and negative increments favor $\theta = 0$. The log-likelihood-ratio scale makes private evidence additive.

Assumption 1. *For every society size n , the private-increment process $\{\Delta_{i,t}\}_{i \in N^{(n)}, t \geq 1}$ satisfies the following conditions with constants $\bar{\Delta}$ and m that do not depend on n :*

- (i) *Conditional on θ , the family $\{\Delta_{i,t}\}_{i \in N^{(n)}, t \geq 1}$ is independent across agents and time.*
- (ii) *There exists $\bar{\Delta} \in (0, \infty)$ such that, for every i and every $t \geq 1$,*

$$|\Delta_{i,t}| \leq \bar{\Delta} \quad a.s.$$

- (iii) *There exists $m > 0$ such that, for every i and every $t \geq 1$,*

$$\mathbb{E}[\Delta_{i,t} \mid \theta = 1] \geq m, \quad \mathbb{E}[\Delta_{i,t} \mid \theta = 0] \leq -m.$$

Bounded private increments rule out one-shot revelation: no single private signal can move a score arbitrarily far in one period. Directional informativeness, the last condition, says that private increments have systematic drift toward the true state.

At the end of period t , agent i updates next period's score according to

$$\ell_{i,t+1} = (1 - \alpha)\ell_{i,t} + \alpha(S_{i,t} + \Delta_{i,t+1}), \quad \alpha \in (0, 1).$$

The parameter α measures responsiveness to current information. A smaller α gives more weight to the inherited score, while a larger α gives more weight to the current social and private inputs. The social aggregate $S_{i,t}$ is a weighted average of neighbors' decoded values, not a weighted average of their latent scores. The private increment $\Delta_{i,t+1}$ then shifts the current input in the direction favored by agent i 's own evidence.

The updating rule is implementable under finite communication. A received message reveals only that a neighbor's latent score lies in a communication bin. Exact Bayesian filtering would require beliefs over neighbors' unobserved score histories, private signals, and higher-order beliefs. The rule above instead uses the objects directly available in every period: the agent's inherited score, decoded values from others, and current private evidence.

2.4 Timing and accuracy

Each period $t \geq 0$ has three stages. First, given current scores $(\ell_{j,t})_{j \in N}$, each agent j sends $m_{j,t} = Q(\ell_{j,t})$, receivers decode $y_{j,t} = \psi(\ell_{j,t})$, and agent i forms $S_{i,t} = \sum_j P_{ij}\psi(\ell_{j,t})$. Second,

agent i observes private evidence summarized by $\Delta_{i,t+1}$. Third, agent i updates

$$\ell_{i,t+1} = (1 - \alpha)\ell_{i,t} + \alpha(S_{i,t} + \Delta_{i,t+1}).$$

Accuracy is measured on the probability scale. A score ℓ induces the probability assessment

$$\sigma(\ell) = \frac{1}{1 + \exp(-\ell)}.$$

Thus agent i 's probability assessment at date t is $a_{i,t} = \sigma(\ell_{i,t})$. Conditional on the realized state θ , define

$$\text{MSE}_t^{(n)}(\theta) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[(a_{i,t} - \theta)^2 \mid \theta].$$

When $\theta = 1$, the term $(a_{i,t} - 1)^2$ vanishes as $\ell_{i,t} \rightarrow \infty$. When $\theta = 0$, the term $a_{i,t}^2$ vanishes as $\ell_{i,t} \rightarrow -\infty$. We study long-run accuracy for fixed societies.

3 Continuous-communication benchmark

We first remove the finite-codebook friction while keeping the same private-increment process and the same score-updating rule. Under continuous communication, agents transmit their real-valued latent scores directly. Formally, replace the finite communication rule with the identity map, $\psi(\ell) = \ell$. The social aggregate vector is then $S_t = P\ell_t$, and the score vector follows

$$\ell_{t+1} = A\ell_t + \alpha\Delta_{t+1}, \quad A = (1 - \alpha)I + \alpha P,$$

where $\ell_t = (\ell_{1,t}, \dots, \ell_{n,t})^\top$ and $\Delta_t = (\Delta_{1,t}, \dots, \Delta_{n,t})^\top$.

For this fixed-society benchmark, suppose that P is irreducible. Since $0 < \alpha < 1$, the matrix $A = (1 - \alpha)I + \alpha P$ is primitive. Since P is row stochastic and irreducible, we have the unique stationary distribution $\pi \in \Delta(N)$ satisfying $\pi^\top P = \pi^\top$. Because A has the same stationary distribution as P , repeated social averaging eventually spreads each agent's private increments throughout the connected society.

Proposition 1. *Maintain Assumption 1, and suppose that P is irreducible. Under continuous communication, for every fixed n and every agent $i \in N$,*

$$\ell_{i,t} \rightarrow +\infty \quad \text{a.s. under } \theta = 1, \quad \ell_{i,t} \rightarrow -\infty \quad \text{a.s. under } \theta = 0.$$

Consequently, $a_{i,t} \rightarrow \theta$ almost surely for every i , and $\text{MSE}_t^{(n)}(\theta) \rightarrow 0$.

Proposition 1 shows that the updating rule itself does not impose an information ceiling. When scores are transmitted continuously, private increments enter future scores, those scores are transmitted to others, and the network repeatedly reallocates accumulated evidence across the society. Hence, despite the reduced-form updating rule, every agent eventually learns the state.

This benchmark clarifies the source of the failure studied below. The obstacle is not the arrival of private evidence, nor the use of social averaging per se. It is the finite communication layer. Once agents transmit decoded messages rather than latent scores, a private increment affects social exposure only when it changes a finite message. The next section studies how this discreteness can turn fast mixing from an aggregating force into an absorbing force.

Figure 1 illustrates the contrast between continuous score transmission and finite message transmission.

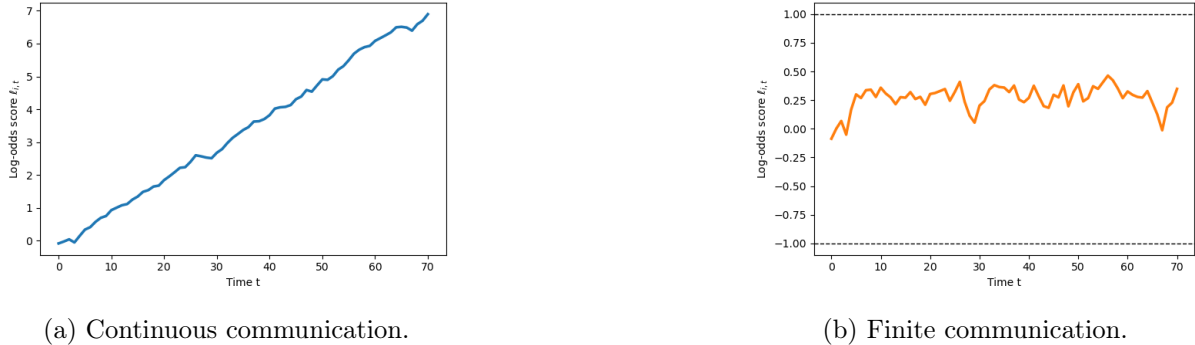


Figure 1: Learning process under continuous and finite communication when $\theta = 1$.

4 Fast Mixing, Buffered Entry, and Same-Message Traps

Under finite communication, the network averages decoded messages rather than latent scores. This creates a new role for mixing. By making agents' social inputs more similar, faster mixing can reduce cross-sectional dispersion in latent scores. If the mean score lies inside a communication bin, this compression can place a large fraction of agents in the same message even though their private scores have not fully incorporated the state.

This section formalizes that mechanism. The first step is a dispersion envelope. The second step turns a buffered mean and low dispersion into same-message concentration. The third step gives conditions under which a large same-message set is absorbing. The result is conditional: mixing is harmful only when the compressed population enters a bin that is robust but insufficiently informative.

4.1 Compression and buffered entry

For $v \in \mathbb{R}^n$, define its empirical mean, centered component, and empirical norm by

$$\bar{v} = \frac{1}{n} \sum_{i=1}^n v_i, \quad \tilde{v} = v - \bar{v} \mathbf{1}, \quad \|v\|_n = \left(\frac{1}{n} \sum_{i=1}^n v_i^2 \right)^{1/2}.$$

Let

$$\Pi = I - \frac{1}{n} \mathbf{1} \mathbf{1}^\top.$$

Then $\Pi v = v - \bar{v} \mathbf{1} = \tilde{v}$. Thus Π subtracts the empirical mean from every coordinate and keeps the part of v that represents cross-sectional dispersion.

For the latent-score vector $\ell_t = (\ell_{1,t}, \dots, \ell_{n,t})^\top$, define

$$D_t = \|\Pi \ell_t\|_n = \left(\frac{1}{n} \sum_{i=1}^n (\ell_{i,t} - \bar{\ell}_t)^2 \right)^{1/2}.$$

Thus D_t is the cross-sectional dispersion of latent scores at date t .

The same centering operation is used after social exposure. If v is a vector of decoded values, then Pv is the vector of exposure-weighted decoded values:

$$(Pv)_i = \sum_{j=1}^n P_{ij} v_j.$$

Applying Π afterward removes the empirical mean of these post-exposure inputs. Therefore ΠPv measures the residual cross-sectional variation that remains after social exposure.

Define the projected mixing coefficient by

$$\lambda(P) = \sup_{\bar{v}=0, v \neq 0} \frac{\|\Pi Pv\|_n}{\|v\|_n}.$$

The coefficient $\lambda(P)$ is the largest possible ratio between residual disagreement and initial disagreement:

$$\|\Pi Pv\|_n \leq \lambda(P) \|v\|_n \quad \text{for every } v \text{ with } \bar{v} = 0.$$

Thus a smaller $\lambda(P)$ means stronger one-step compression of disagreement.

This definition connects directly to the exposure structure. Each row of P describes the population mixture that an agent observes. If different rows of P are similar, then agents receive similar averages of any population-wide disagreement, so Pv is close to a common

vector and $\Pi P v$ is small. If P is segmented, a disagreement vector that is high in one group and low in another can survive exposure, so $\Pi P v$ remains large. In this sense, a lower value of $\lambda(P)$ corresponds to faster one-step mixing.

Let

$$R = \bar{r} - \underline{r}, \quad \underline{r} = r_1, \quad \bar{r} = r_K.$$

Define the long-run dispersion envelope by

$$\bar{D}(P) = \frac{\lambda(P)R}{2} + \bar{\Delta},$$

and, for each date t , define

$$\hat{D}_t(P) = (1 - \alpha)^t D_0 + (1 - (1 - \alpha)^t) \bar{D}(P).$$

Lemma 1. *For every $t \geq 0$, $D_t \leq \hat{D}_t(P)$. Consequently, $\limsup_{t \rightarrow \infty} D_t \leq \bar{D}(P)$.*

Lemma 1 is the compression step. The term $\lambda(P)R/2$ bounds the residual dispersion generated by social exposure, while $\bar{\Delta}$ bounds one-period private noise. Hence a smaller $\lambda(P)$ tightens the range of latent scores around their mean, D_t .

Messages change only at cutpoints. Fix a communication bin $B_k = (b_{k-1}, b_k]$. For $\rho > 0$, define its ρ -buffered interior by

$$I_k(\rho) = (b_{k-1} + \rho, b_k - \rho],$$

with the convention that the lower buffer is ignored when $b_{k-1} = -\infty$ and the upper buffer is ignored when $b_k = +\infty$. When the mean score is close to a cutpoint, small dispersion can still leave many agents on different sides. When the mean score lies safely inside a bin, low dispersion forces most agents to send the same message.

At date t , define the required cutpoint margin for ε -concentration by $\rho_t(P, \varepsilon) = \frac{\hat{D}_t(P)}{\sqrt{\varepsilon}}$. Let

$$C_{k,t} = \{i \in N : \ell_{i,t} \in B_k\}$$

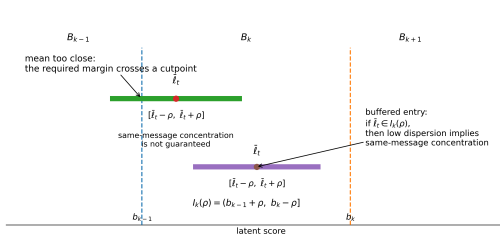
be the set of agents who send message k at date t . Since $D_t \leq \hat{D}_t(P)$, if $\bar{\ell}_t \in I_k(\rho_t(P, \varepsilon))$, then

$$\frac{|C_{k,t}|}{n} \geq 1 - \varepsilon.$$

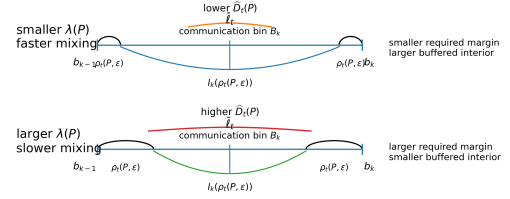
Thus $\rho_t(P, \varepsilon)$ is the cutpoint margin needed to convert the dispersion bound into same-

message concentration.³

The comparative static follows immediately. Holding the other primitives fixed, a smaller projected mixing coefficient $\lambda(P)$ lowers $\hat{D}_t(P)$, and therefore lowers $\rho_t(P, \varepsilon)$. A lower required margin means that less of bin B_k is removed near its cutpoints; equivalently, the buffered interior $I_k(\rho_t(P, \varepsilon))$ becomes larger. Hence, under faster mixing, the mean score need not lie as deeply inside B_k for a large same-message set to form.



(a) Buffered mean for same-message concentration.



(b) Effect of fast mixing on the required margin.

Figure 2: Fast mixing, buffered interiors, and room for the mean.

4.2 Cohesion and robust bins

Same-message concentration is not yet absorption. A large same-message set can still break apart if its members place too much weight on agents outside the set. The set becomes stable only when outside exposure is small enough and the communication bin is robust enough that outside messages and private shocks cannot push its members across a cutpoint.

Define the row-diffuseness coefficient by

$$\omega(P) = n \max_{i,j \in N} P_{ij}.$$

For a sequence of societies $P^{(n)}$, we define bounded row diffuseness, meaning that there exists $C < \infty$, independent of n , such that $\omega(P^{(n)}) \leq C$ for every n . Equivalently, $\max_{i,j} P_{ij}^{(n)} \leq C/n$. Thus no single exposure weight can remain bounded away from zero as the society grows.

The role of row diffuseness is simple. If a set C contains at least a fraction $1 - \varepsilon$ of agents, then every agent's exposure to agents outside C is at most $\omega(P)\varepsilon$. Hence, under bounded row

³The argument is a direct dispersion bound. If more than an ε -fraction of agents were outside B_k , then each such agent would be at distance at least $\rho_t(P, \varepsilon)$ from $\bar{I}_t$. Hence $D_t^2 \geq \varepsilon \rho_t(P, \varepsilon)^2 = \hat{D}_t(P)^2$, contradicting $D_t \leq \hat{D}_t(P)$, up to the weak boundary convention.

diffuseness, large sets are cohesive in large societies: if $\omega(P^{(n)}) \leq C$, then outside exposure is at most $C\varepsilon$.

For $\xi \in [0, 1]$, define the lowest and highest possible social-plus-private input to an agent whose same-message set sends message k and who receives at most ξ outside exposure:

$$\underline{h}_k(\xi) = (1 - \xi)r_k + \xi\underline{r} - \bar{\Delta}, \quad \bar{h}_k(\xi) = (1 - \xi)r_k + \xi\bar{r} + \bar{\Delta}.$$

The same-message part contributes the decoded value r_k . The outside part can be as low as $\underline{r}$ or as high as $\bar{r}$, and the private increment can be as low as $-\bar{\Delta}$ or as high as $\bar{\Delta}$. Thus the current input $S_{i,t} + \Delta_{i,t+1}$ lies between $\underline{h}_k(\xi)$ and $\bar{h}_k(\xi)$.

Definition 1 (Robust bin). The communication bin $B_k = (b_{k-1}, b_k]$ is ξ -robust if

$$b_{k-1} < \underline{h}_k(\xi) \quad \text{whenever } b_{k-1} > -\infty,$$

and

$$\bar{h}_k(\xi) \leq b_k \quad \text{whenever } b_k < +\infty.$$

A bin is ξ -robust when the entire worst-case input interval lies inside the bin, with the same boundary convention as $B_k = (b_{k-1}, b_k]$: strictly above the lower cutpoint and weakly below the upper cutpoint. In that case, if an agent's inherited score is already in B_k , then her next score also remains in B_k , because the update is a convex combination of the inherited score and the current input. Thus outside messages and private shocks cannot push a member of the same-message group across a cutpoint.⁴

4.3 Same-message traps and accuracy loss

We now combine the pieces. Fast mixing bounds dispersion; buffered mean entry turns low dispersion into same-message concentration; diffuse exposure turns concentration into cohesion; and robust bins turn cohesion into absorption.

Theorem 1. Fix a date T , a communication bin B_k , and $\varepsilon \in (0, 1)$. Let

$$\rho_T(P, \varepsilon) = \frac{\hat{D}_T(P)}{\sqrt{\varepsilon}}, \quad \xi(P, \varepsilon) = \min\{\omega(P)\varepsilon, 1\}.$$

Suppose $\bar{\ell}_T \in I_k(\rho_T(P, \varepsilon))$, and suppose B_k is $\xi(P, \varepsilon)$ -robust. Then there exists a set $C_T =$

⁴Since $b_{k-1} \leq \underline{h}_k(\xi) \leq \bar{h}_k(\xi) \leq b_k$, the social aggregation with the private increment value is inside the bin B_k . Therefore, the convex combination with the agent's score and the term will make the updated score in the bin B_k .

$\{i \in N : \ell_{i,T} \in B_k\}$ such that $|C_T|/n \geq 1 - \varepsilon$, and every agent in C_T remains in bin B_k forever:

$$\ell_{i,t} \in B_k \quad \text{and} \quad Q(\ell_{i,t}) = k$$

for every $i \in C_T$ and every $t \geq T$, along every continuation satisfying the private-increment bound in Assumption 1.

Theorem 1 is the general trapping result. It identifies when buffered entry into a robust bin becomes absorbing for a large group. The projected mixing coefficient matters through the required margin: a smaller $\lambda(P)$ lowers the dispersion envelope, lowers $\rho_T(P, \varepsilon)$, and makes same-message concentration easier to obtain.

Same-message absorption matters for accuracy when the absorbing bin is not sufficiently informative about the state. For each communication bin $B_k = (b_{k-1}, b_k]$, define

$$e_k(1) = \begin{cases} (1 - \sigma(b_k))^2, & b_k < +\infty, \\ 0, & b_k = +\infty, \end{cases} \quad e_k(0) = \begin{cases} \sigma(b_{k-1})^2, & b_{k-1} > -\infty, \\ 0, & b_{k-1} = -\infty. \end{cases}$$

Thus $e_k(\theta)$ is the smallest squared error that can be guaranteed when an agent's score remains in bin B_k . We call B_k an interior bin if both of its cutpoints are finite, that is, $b_{k-1} > -\infty$ and $b_k < +\infty$. Equivalently, for the threshold codebook above, the interior bins are $B_2, \dots, B_{K-1}$. For an interior bin, both $e_k(1)$ and $e_k(0)$ are strictly positive.

Corollary 1. *Suppose a set $C \subseteq N$ satisfies $|C|/n \geq 1 - \varepsilon$, and suppose that there exists a date T such that $\ell_{i,t} \in B_k$ for every $i \in C$ and every $t \geq T$. Then, for every $t \geq T$,*

$$\text{MSE}_t^{(n)}(\theta) \geq (1 - \varepsilon)e_k(\theta).$$

In particular, if B_k is an interior bin, then a large trapped core generates a strictly positive long-run error lower bound in both states.

The next section studies which robust bin is selected when the amount of cross-group mixing is varied.

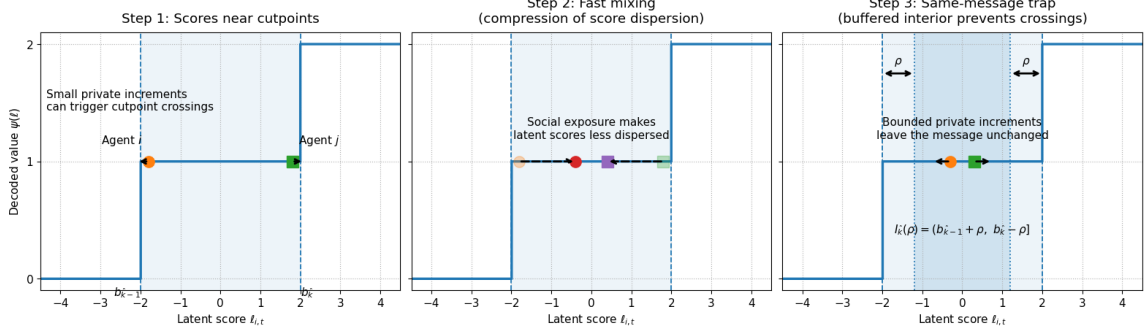


Figure 3: Fast mixing mechanism under finite codebook.

5 Block Mixing and Endogenous Bin Selection

The previous section gives an absorption criterion. Once a large and sufficiently cohesive set enters a robust communication bin, its members keep sending the same message. That result explains persistence, but not selection. It does not say whether the absorbing message is state-aligned, state-opposed, or interior. This section studies selection in a canonical block-mixing society.

Block networks are a standard way to represent homophily, modularity, and group-based exposure. In stochastic block models, agents are partitioned into groups and link probabilities depend on group membership (Holland et al., 1983). In economic models of social networks, group structure and homophily shape who observes whom and how quickly averaging dynamics approach consensus (Currarini et al., 2009; Golub and Jackson, 2012). We use the same reduced-form logic here. The block structure lets the amount of cross-group exposure move one parameter while preserving a transparent two-group geometry.

The block environment isolates two opposing margins. Suppose there are two groups. One group is initially state-aligned, but the other group is not: state lagging. More cross-group exposure helps the lagging group move toward the initially state-aligned group. At the same time, it lowers the initially state-aligned group's exposure to its own message. The phase diagram follows from this tension. Low mixing preserves segmentation, moderate mixing generates state-aligned consensus, and excessive mixing generates non state-aligned consensus.

Throughout this section, fix $\theta = 1$. The case $\theta = 0$ is symmetric after reversing signs and exchanging the low and high messages.

5.1 A canonical block-mixing environment

There are three messages. Fix $c > 0$ and define

$$B_1 = (-\infty, -c], \quad B_2 = (-c, c], \quad B_3 = (c, \infty).$$

The decoded values are symmetric:

$$r_1 = -r, \quad r_2 = 0, \quad r_3 = r,$$

where $r > 0$. Under $\theta = 1$, message 3 is state-aligned, message 1 is state-opposed, and message 2 is interior. We maintain $\bar{\Delta} < c < r - \bar{\Delta}$. The first inequality makes the interior bin robust when all agents send message 2. The second makes the high bin robust when all agents send message 3. By symmetry, the low bin is also robust when all agents send message 1.⁵

Fix a finite society N and partition it into two nonempty groups H and L . Let $|H| \geq 2$, $|L| \geq 2$, and $q = |H|/n \in (1/2, 1)$. Group H is initially state-aligned, while group L is initially lagging:⁶

$$\ell_{i,0} = h_0 > c \quad \text{for } i \in H, \quad \ell_{i,0} = \ell_0^L < -c \quad \text{for } i \in L.$$

Thus H initially sends message 3, and L initially sends message 1. This initial profile displays the full phase diagram in the cleanest form. The mechanism does not require the lagging group to start in the low bin; what matters is that it has not yet crossed into the state-aligned bin. Appendix B.2 records the corresponding case in which the lagging group starts from the interior message.

The parameter $\zeta \in [0, 1]$ measures cross-group mixing. The agent-level exposure matrix P^ζ

⁵The three-message codebook is the minimal codebook that displays the phase diagram in a transparent form. Richer threshold codebooks do not remove the mechanism, but they can change the path by which agents move across bins. If a K -message codebook contains three messages $k_- < k_0 < k_+$ that play the roles of state-opposed, interior, and state-aligned messages, and if the score-containment inequalities used in Theorem 2 hold after replacing B_1, B_2, B_3 and r_1, r_2, r_3 by $B_{k_-}, B_{k_0}, B_{k_+}$ and $r_{k_-}, r_{k_0}, r_{k_+}$, then the same pathwise conclusions follow. Other messages are irrelevant along that path because the maintained inequalities prevent agents from reaching them.

⁶Initial message heterogeneity is essential for the selection problem. If all agents initially send the same robust message, then every agent receives the same decoded social input, independently of ζ . Such a homogeneous profile is absorbing for every mixing parameter, so mixing has no regime-selection role.

has no self-weights and is defined by

$$P_{ij}^\zeta = \begin{cases} \frac{1 - \zeta(1 - q)}{|H| - 1}, & i, j \in H, i \neq j, \\ \frac{\zeta(1 - q)}{|L|}, & i \in H, j \in L, \\ \frac{\zeta q}{|H|}, & i \in L, j \in H, \\ \frac{1 - \zeta q}{|L| - 1}, & i, j \in L, i \neq j, \\ 0, & i = j. \end{cases}$$

Thus an agent in H assigns total weight $1 - \zeta(1 - q)$ to agents in H and total weight $\zeta(1 - q)$ to agents in L . An agent in L assigns total weight ζq to agents in H and total weight $1 - \zeta q$ to agents in L . Since $q > 1/2$, the initially state-aligned group is the larger block.

For block-level calculations, these total exposure weights can be summarized by

$$\mathcal{P}^\zeta = \begin{pmatrix} 1 - \zeta(1 - q) & \zeta(1 - q) \\ \zeta q & 1 - \zeta q \end{pmatrix}.$$

This matrix records the total exposure that an agent in one block assigns to each block. Therefore, if all agents in H send decoded value y_H and all agents in L send decoded value y_L , then

$$\begin{pmatrix} S_H(\zeta; y_H, y_L) \\ S_L(\zeta; y_H, y_L) \end{pmatrix} = \mathcal{P}^\zeta \begin{pmatrix} y_H \\ y_L \end{pmatrix}.$$

Equivalently,

$$S_H(\zeta; y_H, y_L) = (1 - \zeta(1 - q))y_H + \zeta(1 - q)y_L, \quad S_L(\zeta; y_H, y_L) = \zeta q y_H + (1 - \zeta q)y_L.$$

The parameter ζ has a direct exposure interpretation. When $\zeta = 0$, there is no cross-group exposure: agents in H listen only to agents in H , and agents in L listen only to agents in L . As ζ increases, cross-group exposure rises. An H -agent's total exposure to L is $\zeta(1 - q)$, and an L -agent's total exposure to H is ζq . At $\zeta = 1$,

$$\mathcal{P}^1 = \begin{pmatrix} q & 1 - q \\ q & 1 - q \end{pmatrix},$$

so both groups face the same aggregate mixture of H - and L -messages.

This is also the sense in which higher ζ corresponds to faster mixing. As ζ rises, the two block-level exposure rows become closer to each other, so differences between the two groups are averaged away more strongly.⁷ Thus increasing ζ means stronger cross-group averaging in the block environment and stronger one-step compression of cross-sectional dispersion in the sense of Section 4.

Starting from the initial message profile $(3, 1)$, where $y_H = r$ and $y_L = -r$, we obtain

$$\begin{pmatrix} s_H^{31}(\zeta) \\ s_L^{31}(\zeta) \end{pmatrix} = \mathcal{P}^\zeta \begin{pmatrix} r \\ -r \end{pmatrix} = \begin{pmatrix} r(1 - 2(1 - q)\zeta) \\ r(2q\zeta - 1) \end{pmatrix}.$$

The first component is decreasing in ζ : more mixing pulls the initially state-aligned group toward the lagging message. The second component is increasing in ζ : more mixing exposes the lagging group to the state-aligned message. This is the basic selection tradeoff.

5.2 Regime regions and the phase diagram

The selected regime is determined by where reachable scores fall relative to the cutpoints. We use the notation (a, b) for the block message profile in which group H sends message a and group L sends message b . The initial profile is $(3, 1)$. Hence, at date 0, group H sends decoded value r , group L sends decoded value $-r$, and the social aggregates used to form date-one scores are $s_H^{31}(\zeta)$ and $s_L^{31}(\zeta)$.

After one update from the initial profile, the only within-group variation comes from the private increment. Since $|\Delta_{i,1}| \leq \bar{\Delta}$, the possible date-one scores are bounded by

$$I_H^{31}(\zeta) = [\underline{h}_1^{31}(\zeta), \bar{h}_1^{31}(\zeta)], \quad I_L^{31}(\zeta) = [\underline{\ell}_1^{31}(\zeta), \bar{\ell}_1^{31}(\zeta)],$$

where

$$\underline{h}_1^{31}(\zeta) = (1 - \alpha)h_0 + \alpha(s_H^{31}(\zeta) - \bar{\Delta}), \quad \bar{h}_1^{31}(\zeta) = (1 - \alpha)h_0 + \alpha(s_H^{31}(\zeta) + \bar{\Delta}),$$

and

$$\underline{\ell}_1^{31}(\zeta) = (1 - \alpha)\ell_0^L + \alpha(s_L^{31}(\zeta) - \bar{\Delta}), \quad \bar{\ell}_1^{31}(\zeta) = (1 - \alpha)\ell_0^L + \alpha(s_L^{31}(\zeta) + \bar{\Delta}).$$

For a given value of ζ , these intervals, I_H^{31}, I_L^{31} , describe all date-one scores that can arise after one update from the initial message profile $(3, 1)$ for each group.

⁷Appendix B.1 computes the projected mixing coefficient of the full matrix P^ζ and shows that it is weakly decreasing in ζ .

We define three sufficient regime regions.⁸

The segmentation region is

$$\mathcal{Z}_S = \{\zeta \in [0, 1] : s_H^{31}(\zeta) - \bar{\Delta} > c \text{ and } s_L^{31}(\zeta) + \bar{\Delta} < -c\}.$$

For these values of ζ , the initial profile $(3, 1)$ is absorbing. Group H 's worst current input remains above the high cutoff, while group L 's best current input remains below the low cutoff.⁹

Now consider a moderate region where the message path $(3, 1) \rightarrow (3, 2) \rightarrow (3, 3)$ is guaranteed. The first step requires

$$I_H^{31}(\zeta) \subset (c, \infty) \quad \text{and} \quad I_L^{31}(\zeta) \subset (-c, c).$$

These containments mean that all agents in H remain in message 3 at date one, while all agents in L move from message 1 into message 2. Given the intermediate profile $(3, 2)$, the block social aggregates are

$$s_H^{32}(\zeta) = r(1 - \zeta(1 - q)), \quad s_L^{32}(\zeta) = \zeta qr.$$

To guarantee that all agents enter the high bin at date two, it is enough to bound date-two scores from below. Define

$$\underline{h}_2^M(\zeta) = (1 - \alpha)\underline{h}_1^{31}(\zeta) + \alpha(s_H^{32}(\zeta) - \bar{\Delta}), \quad \underline{\ell}_2^M(\zeta) = (1 - \alpha)\underline{\ell}_1^{31}(\zeta) + \alpha(s_L^{32}(\zeta) - \bar{\Delta}).$$

The moderate region is

$$\mathcal{Z}_M = \{\zeta \in [0, 1] : I_H^{31}(\zeta) \subset (c, \infty), I_L^{31}(\zeta) \subset (-c, c), \underline{h}_2^M(\zeta) > c, \underline{\ell}_2^M(\zeta) > c\}.$$

The interior-consensus region is

$$\mathcal{Z}_I = \{\zeta \in [0, 1] : I_H^{31}(\zeta) \subset (-c, c) \text{ and } I_L^{31}(\zeta) \subset (-c, c)\}.$$

For these values of ζ , both groups enter the interior bin after the first update. Since $\bar{\Delta} < c$, the interior bin is robust when all agents send message 2.

⁸They are not an exhaustive partition of all parameter values. Other mixed absorbing profiles may arise outside them. The point is to isolate the ordered regimes that generate the nonmonotone value of mixing.

⁹Since $h_0 > c$ and $s_H^{31}(\zeta) - \bar{\Delta} > c$, the update keeps every H -agent above c . The argument for L is symmetric.

Theorem 2. Consider the block-mixing environment above and maintain $\bar{\Delta} < c < r - \bar{\Delta}$. Under Assumption 1, the following statements hold.

1. The regime regions are ordered by the mixing parameter. If $\zeta_S \in \mathcal{Z}_S$, $\zeta_M \in \mathcal{Z}_M$, and $\zeta_I \in \mathcal{Z}_I$, then $\zeta_S < \zeta_M < \zeta_I$.
2. If $\zeta \in \mathcal{Z}_S$, then the initial profile $(3, 1)$ is absorbing. Group H sends message 3 forever, and group L sends message 1 forever.
3. If $\zeta \in \mathcal{Z}_M$, then the path $(3, 1) \rightarrow (3, 2) \rightarrow (3, 3)$ is guaranteed pathwise, and all agents send message 3 from date 2 onward.
4. If $\zeta \in \mathcal{Z}_I$, then the path $(3, 1) \rightarrow (2, 2)$ is guaranteed pathwise, and all agents send message 2 from date 1 onward.

The theorem turns the ordered regions into a pathwise selection result. Moving from low to intermediate mixing changes the selected regime from segmentation to state-aligned consensus. Moving from intermediate to high mixing changes the selected regime from state-aligned consensus to interior consensus.

The theorem should be read as a sufficient phase diagram for the canonical heterogeneous initial profile. Homogeneous robust profiles, such as $(3, 3)$ or $(2, 2)$, are already absorbing for every ζ , so mixing has no regime-selection role there. The relevant object is initial message heterogeneity: one group begins with a state-aligned message, while another group has not yet crossed into that message. Appendix B.2 shows that the same selection force also arises when the lagging group starts from the interior message $(3, 2)$.

The phase diagram has an immediate accuracy implication. Write $\text{MSE}_t^{(n)}(1; \zeta)$ for the mean-squared error generated by the block-mixing matrix P^ζ , conditional on $\theta = 1$. Under $\theta = 1$, the high message r_3 is state-aligned. The interior message r_2 is less informative because it keeps scores from crossing the high cutoff. Hence moderate mixing is more accurate than excessive mixing because the former selects the high bin, while the latter selects the interior bin.

Corollary 2. Suppose $\mathcal{Z}_M$ and $\mathcal{Z}_I$ are nonempty. If $\zeta_M \in \mathcal{Z}_M$ and $\zeta_I \in \mathcal{Z}_I$, then, conditional on $\theta = 1$,

$$\text{MSE}_t^{(n)}(1; \zeta_M) < \text{MSE}_t^{(n)}(1; \zeta_I) \quad \text{for every } t \geq 2.$$

If, in addition, $\mathcal{Z}_S$ is nonempty and q is sufficiently small so that $(1 - \sigma(c))^2 < (1 - q)\sigma(c)^2$, then, for every $\zeta_M \in \mathcal{Z}_M$ and every $\zeta_S \in \mathcal{Z}_S$,

$$\text{MSE}_t^{(n)}(1; \zeta_M) < \text{MSE}_t^{(n)}(1; \zeta_S) \quad \text{for every } t \geq 2.$$

The first comparison is direct. Under $\zeta_M \in \mathcal{Z}_M$, every agent's score is above c from date 2 onward. Therefore every agent's squared error is strictly below $(1 - \sigma(c))^2$. Under $\zeta_I \in \mathcal{Z}_I$, every agent's score is at most c from date 1 onward. Therefore every agent's squared error is at least $(1 - \sigma(c))^2$. Moderate mixing therefore strictly dominates excessive mixing.

The comparison with segmentation requires one extra sufficient condition because segmentation traps only the smaller group L in the low bin. Under $\zeta_S \in \mathcal{Z}_S$, all agents in L have scores at most $-c$, so their squared error is at least $(1 - \sigma(-c))^2 = \sigma(c)^2$. Since the mass of this group is $1 - q$, segmentation has loss at least $(1 - q)\sigma(c)^2$. The displayed condition therefore guarantees that moderate mixing also strictly dominates low-mixing segmentation.

The value of mixing is therefore nonmonotone. Moving from low to moderate mixing helps because the initially state-aligned group pulls the lagging group into the high bin. Moving from moderate to excessive mixing hurts because cross-group averaging becomes strong enough to pull the initially state-aligned group into the interior bin before high-message consensus is reached.

6 Conclusion

This paper studies social learning when agents communicate through finite messages rather than real-valued assessments. Finite communication creates a transmission problem: information can continue to accumulate privately while remaining invisible to the network.

The main result is that social mixing has a nonmonotone value under this friction. Mixing helps agents learn from others, but it also compresses latent disagreement. When messages are finite, this compression can be costly because message changes require cutpoint crossings. If mixing pulls many agents into the same communication bin before enough state-aligned crossings occur, the society can become trapped in a common message. The trap is absorbing when the bin is robust to private shocks and outside exposure. If the absorbing bin is an interior bin, the result is persistent error even though agents continue to receive informative private evidence.

The block-mixing environment makes this force transparent. When cross-group exposure is low, the initially different groups remain segmented. When cross-group exposure is moderate, the state-aligned group pulls the lagging group into the state-aligned message. When cross-group exposure is excessive, both groups are pulled into an interior message before state-aligned consensus forms. Thus the same increase in mixing that first improves learning eventually worsens it. Better connectivity need not improve learning when the communication system is too coarse.

The paper therefore separates two roles that are often bundled together in models of social learning. Averaging helps aggregate information, but under finite communication, information is socially transmitted only through message changes. With real-valued communication these two forces move together: disagreement can be averaged while the underlying assessments remain visible. With finite communication they can come apart: averaging can reduce the very disagreement that would have produced informative message changes.

This mechanism suggests a simple design lesson. Communication systems and network structures should be considered jointly. Coarse messages may be harmless in sparse or weakly mixed environments, where disagreement has room to generate threshold crossings. The same messages may be harmful in highly mixed environments, where averaging quickly compresses assessments into stable intermediate categories. In this sense, stronger social connectivity raises the value of communication resolution. When agents are more tightly connected, the message system must be rich enough to keep private learning socially visible.

A Proofs

A.1 Proofs of main-text results

Proof of Proposition 1

Under continuous communication, the score vector satisfies

$$\ell_t = A^t \ell_0 + \alpha \sum_{s=1}^t A^{t-s} \Delta_s, \quad A = (1 - \alpha)I + \alpha P.$$

Since P is irreducible and $0 < \alpha < 1$, A is primitive. Let π be the unique stationary distribution of P . Then $\pi^\top A = \pi^\top$, and $A^k \rightarrow \mathbf{1}\pi^\top$. Since A is primitive, there exist constants $C < \infty$ and $\rho \in (0, 1)$ such that

$$\|A^k - \mathbf{1}\pi^\top\| \leq C\rho^k \quad \text{for every } k \geq 0.$$

Fix $\theta = 1$. Let $Z_s = \pi^\top \Delta_s$. By Assumption 1, the variables $\{Z_s\}_{s \geq 1}$ are conditionally independent under $\theta = 1$ and uniformly bounded. Moreover,

$$\mathbb{E}[Z_s \mid \theta = 1] = \sum_{i=1}^n \pi_i \mathbb{E}[\Delta_{i,s} \mid \theta = 1] \geq m \sum_{i=1}^n \pi_i = m.$$

The strong law for independent uniformly bounded variables gives

$$\frac{1}{t} \sum_{s=1}^t (Z_s - \mathbb{E}[Z_s \mid \theta = 1]) \rightarrow 0 \quad \text{a.s. under } \theta = 1.$$

Hence

$$\sum_{s=1}^t \pi^\top \Delta_s = \sum_{s=1}^t Z_s \geq mt + o(t) \rightarrow +\infty \quad \text{a.s. under } \theta = 1.$$

It remains to show that replacing A^{t-s} by $\mathbf{1}\pi^\top$ changes the accumulated evidence by only a bounded term. Since the increments are uniformly bounded, there is $M < \infty$ such that $\|\Delta_s\| \leq M$ for every s . Therefore

$$\left\| \sum_{s=1}^t (A^{t-s} - \mathbf{1}\pi^\top) \Delta_s \right\| \leq \sum_{s=1}^t C\rho^{t-s}M \leq \frac{CM}{1-\rho}.$$

Thus

$$\sum_{s=1}^t A^{t-s} \Delta_s = \mathbf{1} \sum_{s=1}^t \pi^\top \Delta_s + O(1),$$

where the $O(1)$ term is uniform in t . Since $A^t \ell_0 \rightarrow \mathbf{1}\pi^\top \ell_0$, the initial-condition term is also bounded. Consequently, each coordinate of ℓ_t diverges to $+\infty$ almost surely under $\theta = 1$.

The proof under $\theta = 0$ is symmetric. In that case $\mathbb{E}[\pi^\top \Delta_s \mid \theta = 0] \leq -m$, so the same strong-law argument gives

$$\sum_{s=1}^t \pi^\top \Delta_s \rightarrow -\infty \quad \text{a.s. under } \theta = 0.$$

The same geometric bound then implies $\ell_{i,t} \rightarrow -\infty$ almost surely for every i . Since $\sigma(\ell) \rightarrow 1$ as $\ell \rightarrow +\infty$ and $\sigma(\ell) \rightarrow 0$ as $\ell \rightarrow -\infty$, we have $a_{i,t} \rightarrow \theta$ almost surely for every i . Bounded convergence then implies $\text{MSE}_t^{(n)}(\theta) \rightarrow 0$. ■

Proof of Lemma 1

Let $y_t = (\psi(\ell_{1,t}), \dots, \psi(\ell_{n,t}))^\top$. The score recursion is

$$\ell_{t+1} = (1 - \alpha)\ell_t + \alpha P y_t + \alpha \Delta_{t+1}.$$

Applying Π and using the triangle inequality gives

$$D_{t+1} \leq (1 - \alpha)D_t + \alpha\|\Pi Py_t\|_n + \alpha\|\Pi\Delta_{t+1}\|_n.$$

Because $P\mathbf{1} = \mathbf{1}$, we have $\Pi Py_t = \Pi P \Pi y_t$. By the definition of $\lambda(P)$, $\|\Pi Py_t\|_n \leq \lambda(P)\|\Pi y_t\|_n$. Since decoded values lie in $[\underline{r}, \bar{r}]$, Lemma 2 gives $\|\Pi y_t\|_n \leq R/2$. Since $|\Delta_{i,t+1}| \leq \bar{\Delta}$, we also have $\|\Pi\Delta_{t+1}\|_n \leq \bar{\Delta}$. Therefore

$$D_{t+1} \leq (1 - \alpha)D_t + \alpha \left(\frac{\lambda(P)R}{2} + \bar{\Delta} \right) = (1 - \alpha)D_t + \alpha \bar{D}(P).$$

Iterating this recursion gives

$$D_t \leq (1 - \alpha)^t D_0 + (1 - (1 - \alpha)^t) \bar{D}(P) = \hat{D}_t(P).$$

Since $\hat{D}_t(P) \rightarrow \bar{D}(P)$, it follows that $\limsup_{t \rightarrow \infty} D_t \leq \bar{D}(P)$. ■

Proof of Theorem 1

By Lemma 1, $D_T \leq \hat{D}_T(P)$. Since $\bar{\ell}_T \in I_k(\hat{D}_T(P)/\sqrt{\varepsilon})$, Lemma 3 implies that

$$C_T = \{i \in N : \ell_{i,T} \in B_k\}$$

satisfies $|C_T|/n \geq 1 - \varepsilon$. By Lemma 4,

$$\sum_{j \notin C_T} P_{ij} \leq \omega(P)\varepsilon \quad \text{for every } i \in N.$$

Since total outside exposure is also at most one,

$$\sum_{j \notin C_T} P_{ij} \leq \min\{\omega(P)\varepsilon, 1\} = \xi(P, \varepsilon) \quad \text{for every } i \in C_T.$$

Because B_k is $\xi(P, \varepsilon)$ -robust, Proposition 2 implies that every agent in C_T remains in B_k and sends message k at every date $t \geq T$. ■

Proof of Corollary 1

Suppose first that $\theta = 1$. If $\ell_{i,t} \in B_k$, then $\ell_{i,t} \leq b_k$ whenever $b_k < +\infty$. Hence $a_{i,t} = \sigma(\ell_{i,t}) \leq \sigma(b_k)$, so $(a_{i,t} - 1)^2 \geq (1 - \sigma(b_k))^2 = e_k(1)$. If $b_k = +\infty$, then $e_k(1) = 0$, and the inequality is trivial. Now suppose $\theta = 0$. If $\ell_{i,t} \in B_k$, then $\ell_{i,t} > b_{k-1}$ whenever $b_{k-1} > -\infty$.

Hence $a_{i,t} = \sigma(\ell_{i,t}) > \sigma(b_{k-1})$, so $a_{i,t}^2 \geq \sigma(b_{k-1})^2 = e_k(0)$. If $b_{k-1} = -\infty$, then $e_k(0) = 0$, and the inequality is trivial. Therefore each trapped agent contributes at least $e_k(\theta)$ to squared error in state θ . Since at least a fraction $1 - \varepsilon$ of agents are in C ,

$$\text{MSE}_t^{(n)}(\theta) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[(a_{i,t} - \theta)^2 \mid \theta] \geq (1 - \varepsilon)e_k(\theta).$$

■

Proof of Theorem 2

The ordering statement follows from Lemma 6.

Fix $\zeta \in \mathcal{Z}_S$. Under the initial profile $(3, 1)$, every agent in H receives social aggregate $s_H^{31}(\zeta)$, and every agent in L receives social aggregate $s_L^{31}(\zeta)$. The definition of $\mathcal{Z}_S$ gives $s_H^{31}(\zeta) - \bar{\Delta} > c$ and $s_L^{31}(\zeta) + \bar{\Delta} < -c$. Thus every admissible current input of an H -agent is above c , while every admissible current input of an L -agent is below $-c$. Since $h_0 > c$ and $\ell_0^L < -c$, the update rule implies that all H -agents remain in B_3 and all L -agents remain in B_1 after one update. The same argument repeats at every later date, so $(3, 1)$ is absorbing.

Next fix $\zeta \in \mathcal{Z}_M$. By the first two requirements in the definition of $\mathcal{Z}_M$, after one update from $(3, 1)$, every H -agent remains in B_3 and every L -agent enters B_2 . Hence the date-one message profile is $(3, 2)$. Given this profile, the block social aggregates are $s_H^{32}(\zeta)$ and $s_L^{32}(\zeta)$. For every H -agent, the date-two score is bounded below by $\underline{h}_2^M(\zeta)$, and for every L -agent, the date-two score is bounded below by $\underline{\ell}_2^M(\zeta)$. The definition of $\mathcal{Z}_M$ gives $\underline{h}_2^M(\zeta) > c$ and $\underline{\ell}_2^M(\zeta) > c$. Therefore all agents send message 3 from date 2 onward. Lemma 5 then implies that this high-message consensus is absorbing.

Finally fix $\zeta \in \mathcal{Z}_I$. By the definition of $\mathcal{Z}_I$, after one update from $(3, 1)$, every agent in both groups has a score in B_2 . Hence the date-one message profile is $(2, 2)$. Since $\bar{\Delta} < c$, Lemma 5 implies that interior consensus is absorbing. Thus all agents send message 2 from date 1 onward. ■

Proof of Corollary 2

Fix $\zeta_M \in \mathcal{Z}_M$ and $\zeta_I \in \mathcal{Z}_I$. By Theorem 2, under ζ_M , every score is above c from date 2 onward. Hence, conditional on $\theta = 1$, every squared error is strictly below $(1 - \sigma(c))^2$. Under ζ_I , every score is at most c from date 1 onward. Hence every squared error is at least $(1 - \sigma(c))^2$. Therefore

$$\text{MSE}_t^{(n)}(1; \zeta_M) < \text{MSE}_t^{(n)}(1; \zeta_I) \quad \text{for every } t \geq 2.$$

Now fix $\zeta_S \in \mathcal{Z}_S$. By Theorem 2, every agent in L remains in B_1 , so each such agent has score at most $-c$. Conditional on $\theta = 1$, each L -agent's squared error is therefore at least $(1 - \sigma(-c))^2 = \sigma(c)^2$. Since the mass of group L is $1 - q$,

$$\text{MSE}_t^{(n)}(1; \zeta_S) \geq (1 - q)\sigma(c)^2 \quad \text{for every } t \geq 0.$$

Under ζ_M , the first part of the proof gives

$$\text{MSE}_t^{(n)}(1; \zeta_M) < (1 - \sigma(c))^2 \quad \text{for every } t \geq 2.$$

Thus, if $(1 - \sigma(c))^2 < (1 - q)\sigma(c)^2$, then

$$\text{MSE}_t^{(n)}(1; \zeta_M) < \text{MSE}_t^{(n)}(1; \zeta_S) \quad \text{for every } t \geq 2.$$

■

A.2 Auxiliary results

Lemma 2. *If $z \in [\underline{z}, \bar{z}]^n$, then*

$$\|\Pi z\|_n \leq \frac{\bar{z} - \underline{z}}{2}.$$

Proof of Lemma 2

The empirical variance of a bounded variable supported on an interval of length $\bar{z} - \underline{z}$ is at most $(\bar{z} - \underline{z})^2/4$. Since $\|\Pi z\|_n^2$ is the empirical variance of z , the result follows. ■

Lemma 3. *Fix $\varepsilon \in (0, 1)$, a date t , and $D \geq 0$. Suppose $D_t \leq D$ and $\bar{\ell}_t \in I_k(D/\sqrt{\varepsilon})$. Then at least a fraction $1 - \varepsilon$ of agents send message k at date t :*

$$\frac{1}{n} |\{i \in N : \ell_{i,t} \in B_k\}| \geq 1 - \varepsilon.$$

Proof of Lemma 3

If $D = 0$, then $D_t = 0$, so all scores equal $\bar{\ell}_t$. Since $\bar{\ell}_t \in I_k(0) = B_k$, all agents send message k . If $D > 0$, the argument below applies.

Let $A = \{i \in N : \ell_{i,t} \notin B_k\}$. If $i \in A$, then either $\ell_{i,t} \leq b_{k-1}$ or $\ell_{i,t} > b_k$. Since $\bar{\ell}_t \in I_k(D/\sqrt{\varepsilon})$, every such agent is at distance at least $D/\sqrt{\varepsilon}$ from the mean. Hence

$$D_t^2 = \frac{1}{n} \sum_{i=1}^n (\ell_{i,t} - \bar{\ell}_t)^2 \geq \frac{|A|}{n} \frac{D^2}{\varepsilon}.$$

Since $D_t \leq D$, this implies $|A|/n \leq \varepsilon$. Therefore at least a fraction $1 - \varepsilon$ of agents lie in B_k and send message k . ■

Lemma 4. *Let $C \subseteq N$ satisfy $|C|/n \geq 1 - \varepsilon$. Then, for every $i \in N$,*

$$\sum_{j \notin C} P_{ij} \leq \omega(P)\varepsilon.$$

Proof of Lemma 4

Since $|N \setminus C| \leq \varepsilon n$,

$$\sum_{j \notin C} P_{ij} \leq |N \setminus C| \max_{i,j} P_{ij} \leq \varepsilon n \max_{i,j} P_{ij} = \omega(P)\varepsilon.$$

■

Proposition 2. *Fix a set $C \subseteq N$, a message k , and $\xi \in [0, 1]$. Suppose*

$$\sum_{j \notin C} P_{ij} \leq \xi \quad \text{for every } i \in C,$$

and suppose B_k is ξ -robust. If, at some date T , $\ell_{i,T} \in B_k$ for every $i \in C$, then, along every realization of future private increments satisfying $|\Delta_{i,s}| \leq \bar{\Delta}$, we have

$$\ell_{i,t} \in B_k \quad \text{and} \quad Q(\ell_{i,t}) = k$$

for every $i \in C$ and every $t \geq T$.

Proof of Proposition 2

The proof is by induction. The claim holds at date T by assumption. Suppose it holds at date $t \geq T$. Then every agent in C sends message k at date t . Fix $i \in C$, and write

$$\eta_i = \sum_{j \notin C} P_{ij}.$$

By assumption, $\eta_i \leq \xi$. Since every agent in C sends message k , the social aggregate can be written as

$$S_{i,t} = (1 - \eta_i)r_k + \sum_{j \notin C} P_{ij}\psi(\ell_{j,t}).$$

If $\eta_i > 0$, define

$$x_{i,t} = \frac{1}{\eta_i} \sum_{j \notin C} P_{ij} \psi(\ell_{j,t}).$$

Then $x_{i,t} \in [\underline{r}, \bar{r}]$, because all decoded values lie in $[\underline{r}, \bar{r}]$, and

$$S_{i,t} = (1 - \eta_i)r_k + \eta_i x_{i,t}.$$

If $\eta_i = 0$, the same representation holds after choosing any $x_{i,t} \in [\underline{r}, \bar{r}]$.

Because $r_k \in [\underline{r}, \bar{r}]$, $x_{i,t} \in [\underline{r}, \bar{r}]$, and $\eta_i \leq \xi$, the lowest possible value of $S_{i,t}$ is obtained by placing outside exposure on $\underline{r}$, and the highest possible value is obtained by placing outside exposure on $\bar{r}$. Hence

$$(1 - \xi)r_k + \xi\underline{r} \leq S_{i,t} \leq (1 - \xi)r_k + \xi\bar{r}.$$

Using $|\Delta_{i,t+1}| \leq \bar{\Delta}$, we obtain

$$\underline{h}_k(\xi) \leq S_{i,t} + \Delta_{i,t+1} \leq \bar{h}_k(\xi).$$

Since B_k is ξ -robust, the current input $S_{i,t} + \Delta_{i,t+1}$ lies in B_k . By the induction hypothesis, $\ell_{i,t} \in B_k$. The update

$$\ell_{i,t+1} = (1 - \alpha)\ell_{i,t} + \alpha(S_{i,t} + \Delta_{i,t+1})$$

is a convex combination of two points in $B_k = (b_{k-1}, b_k]$, and therefore also lies in B_k . Hence $Q(\ell_{i,t+1}) = k$. This completes the induction. $\blacksquare$

Lemma 5. *Maintain $\bar{\Delta} < c < r - \bar{\Delta}$. If all agents send message 3 at some date, then every agent who has a score in B_3 remains in B_3 at the next date. If all agents send message 2 at some date, then every agent who has a score in B_2 remains in B_2 at the next date. If all agents send message 1 at some date, then every agent who has a score in B_1 remains in B_1 at the next date.*

Proof of Lemma 5

If all agents send message 3, then $S_{i,t} = r$ for every i . Hence $S_{i,t} + \Delta_{i,t+1} \geq r - \bar{\Delta} > c$. If $\ell_{i,t} > c$, then $\ell_{i,t+1} = (1 - \alpha)\ell_{i,t} + \alpha(S_{i,t} + \Delta_{i,t+1}) > c$. The proof for message 1 is symmetric. If all agents send message 2, then $S_{i,t} = 0$ for every i , so $-\bar{\Delta} \leq S_{i,t} + \Delta_{i,t+1} \leq \bar{\Delta}$. Since $\bar{\Delta} < c$, this current input lies in $(-c, c)$. A convex combination of a point in $B_2 = (-c, c]$ and a point in $(-c, c)$ remains in B_2 . $\blacksquare$

Lemma 6. *If $\zeta_S \in \mathcal{Z}_S$, $\zeta_M \in \mathcal{Z}_M$, and $\zeta_I \in \mathcal{Z}_I$, then*

$$\zeta_S < \zeta_M < \zeta_I.$$

Proof of Lemma 6

The function $s_L^{31}(\zeta) = r(2q\zeta - 1)$ is strictly increasing in ζ . If $\zeta_S \in \mathcal{Z}_S$, then $s_L^{31}(\zeta_S) + \bar{\Delta} < -c$. If $\zeta_M \in \mathcal{Z}_M$, then $\underline{\ell}_1^{31}(\zeta_M) > -c$. Since $\ell_0^L < -c$, the inequality

$$(1 - \alpha)\ell_0^L + \alpha(s_L^{31}(\zeta_M) - \bar{\Delta}) > -c$$

implies $s_L^{31}(\zeta_M) - \bar{\Delta} > -c$. Hence $s_L^{31}(\zeta_M) > s_L^{31}(\zeta_S)$, and strict monotonicity gives $\zeta_S < \zeta_M$.

Next, the function $\underline{h}_1^{31}(\zeta) = (1 - \alpha)h_0 + \alpha(s_H^{31}(\zeta) - \bar{\Delta})$ is strictly decreasing in ζ , because $s_H^{31}(\zeta) = r(1 - 2(1 - q)\zeta)$ is strictly decreasing. If $\zeta_M \in \mathcal{Z}_M$, then $\underline{h}_1^{31}(\zeta_M) > c$. If $\zeta_I \in \mathcal{Z}_I$, then $\bar{h}_1^{31}(\zeta_I) \leq c$, and hence $\underline{h}_1^{31}(\zeta_I) < c$. Therefore $\underline{h}_1^{31}(\zeta_M) > \underline{h}_1^{31}(\zeta_I)$, so strict monotonicity gives $\zeta_M < \zeta_I$. ■

B Robustness and Details for the Block-Mixing Section

This appendix collects two details for Section 5. First, we compute the projected mixing coefficient of the block matrix P^ζ . This connects the block parameter ζ to the general compression measure used in Section 4. Second, we show that the same selection force arises when the lagging group starts from the interior message rather than the low message.

B.1 Projected mixing coefficient of the block matrix

Let $h = |H|$, $l = |L|$, and $q = h/n$. Recall that $h \geq 2$, $l \geq 2$, and $q \in (1/2, 1)$. The agent-level block matrix P^ζ is symmetric. Within each block all off-diagonal weights are equal, and for $i \in H$, $j \in L$,

$$P_{ij}^\zeta = \frac{\zeta(1 - q)}{|L|} = \frac{\zeta}{n} = \frac{\zeta q}{|H|} = P_{ji}^\zeta.$$

Since P^ζ is symmetric and row-stochastic, it is doubly stochastic. Moreover, symmetry makes P^ζ self-adjoint under the empirical inner product $\langle u, v \rangle_n = n^{-1} \sum_i u_i v_i$. Hence the projected mixing coefficient is the largest absolute nonconstant eigenvalue of P^ζ .

Let

$$\lambda(P^\zeta) = \sup_{\bar{v}=0, v \neq 0} \frac{\|\Pi P^\zeta v\|_n}{\|v\|_n}.$$

Since P^ζ is doubly stochastic, it maps mean-zero vectors into mean-zero vectors. Therefore, on the mean-zero subspace, $\Pi P^\zeta = P^\zeta$.

Lemma 7. *For the block matrix P^ζ ,*

$$\lambda(P^\zeta) = \max \left\{ 1 - \zeta, \frac{1 - \zeta(1 - q)}{|H| - 1}, \frac{1 - \zeta q}{|L| - 1} \right\}.$$

In particular, $\lambda(P^\zeta)$ is weakly decreasing in ζ .

Proof of Lemma 7

Since P^ζ is symmetric and row-stochastic, it is doubly stochastic. Hence it preserves the mean-zero subspace, and on that subspace $\Pi P^\zeta = P^\zeta$. Moreover, symmetry implies that P^ζ is self-adjoint under the empirical inner product. Therefore $\lambda(P^\zeta)$ is the largest absolute eigenvalue of P^ζ on the mean-zero subspace.

Decompose $\mathbb{R}^n$ into the mutually orthogonal subspaces spanned by $\mathbf{1}$, the block-contrast subspace $E_B = \{v : v_i = x \text{ for } i \in H, v_i = y \text{ for } i \in L, qx + (1 - q)y = 0\}$, the within- H subspace $E_H = \{v : v_i = 0 \text{ for } i \in L, \sum_{i \in H} v_i = 0\}$, and the within- L subspace $E_L = \{v : v_i = 0 \text{ for } i \in H, \sum_{i \in L} v_i = 0\}$. These subspaces are invariant under P^ζ .

The constant subspace has eigenvalue 1, which is removed by restricting to mean-zero vectors. Next take $v \in E_B$, so $v_i = x$ on H , $v_i = y$ on L , and $qx + (1 - q)y = 0$. For $i \in H$,

$$(P^\zeta v)_i = (1 - \zeta(1 - q))x + \zeta(1 - q)y = x + \zeta(1 - q)(y - x).$$

Since $qx + (1 - q)y = 0$, we have $(1 - q)(y - x) = -x$. Hence $(P^\zeta v)_i = (1 - \zeta)x$. Similarly, for $i \in L$,

$$(P^\zeta v)_i = \zeta qx + (1 - \zeta q)y = y + \zeta q(x - y),$$

and $q(x - y) = -y$, so $(P^\zeta v)_i = (1 - \zeta)y$. Thus E_B is an eigenspace with eigenvalue $1 - \zeta$.

Now take $v \in E_H$. For $i \in H$,

$$(P^\zeta v)_i = \frac{1 - \zeta(1 - q)}{|H| - 1} \sum_{j \in H, j \neq i} v_j = -\frac{1 - \zeta(1 - q)}{|H| - 1} v_i,$$

because $\sum_{j \in H} v_j = 0$. For $i \in L$, $(P^\zeta v)_i = 0$. Hence E_H is an eigenspace with eigenvalue

$$-\frac{1 - \zeta(1 - q)}{|H| - 1}.$$

The same argument on E_L gives eigenvalue

$$-\frac{1 - \zeta q}{|L| - 1}.$$

Since P^ζ is self-adjoint and the mean-zero subspace is the orthogonal direct sum of E_B , E_H , and E_L , the operator norm of P^ζ on the mean-zero subspace is the maximum absolute value of these three eigenvalues. Therefore

$$\lambda(P^\zeta) = \max \left\{ 1 - \zeta, \frac{1 - \zeta(1 - q)}{|H| - 1}, \frac{1 - \zeta q}{|L| - 1} \right\}.$$

Each term inside the maximum is weakly decreasing in ζ , so $\lambda(P^\zeta)$ is weakly decreasing in ζ . ■

Lemma 7 gives the formal connection between the block parameter ζ and the general mixing measure used in Section 4. The term $1 - \zeta$ is the block-level contraction. The remaining two terms are within-block residual terms generated by the no-self-weight convention. They vanish as the corresponding block sizes grow and are also weakly decreasing in ζ .

B.2 The initial profile (3, 2)

The main text starts from (3, 1), where the initially state-aligned group sends the high message and the lagging group sends the low message. The profile (3, 2) shows that the selection force does not require the lagging group to begin at the bottom of the codebook. What matters is that one group has already crossed into the state-aligned bin while the other has not.

Assume now that

$$\ell_{i,0} = h_0 > c \quad \text{for } i \in H, \quad \ell_{i,0} = m_0 \in (-c, c) \quad \text{for } i \in L.$$

The initial message profile is (3, 2). The block social inputs are

$$s_H^{32}(\zeta) = r(1 - \zeta(1 - q)), \quad s_L^{32}(\zeta) = \zeta q r.$$

After one update from the profile $(3, 2)$, the possible date-one scores are contained in

$$I_H^{32}(\zeta) = [\underline{h}_1^{32}(\zeta), \bar{h}_1^{32}(\zeta)], \quad I_L^{32}(\zeta) = [\underline{\ell}_1^{32}(\zeta), \bar{\ell}_1^{32}(\zeta)],$$

where

$$\underline{h}_1^{32}(\zeta) = (1 - \alpha)h_0 + \alpha(s_H^{32}(\zeta) - \bar{\Delta}), \quad \bar{h}_1^{32}(\zeta) = (1 - \alpha)h_0 + \alpha(s_H^{32}(\zeta) + \bar{\Delta}),$$

and

$$\underline{\ell}_1^{32}(\zeta) = (1 - \alpha)m_0 + \alpha(s_L^{32}(\zeta) - \bar{\Delta}), \quad \bar{\ell}_1^{32}(\zeta) = (1 - \alpha)m_0 + \alpha(s_L^{32}(\zeta) + \bar{\Delta}).$$

Define three sufficient outcome regions:

$$\mathcal{Z}_A^{32} = \{\zeta \in [0, 1] : s_H^{32}(\zeta) - \bar{\Delta} > c, -c < s_L^{32}(\zeta) - \bar{\Delta}, s_L^{32}(\zeta) + \bar{\Delta} < c\},$$

$$\mathcal{Z}_H^{32} = \{\zeta \in [0, 1] : I_H^{32}(\zeta) \subset (c, \infty), I_L^{32}(\zeta) \subset (c, \infty)\},$$

and

$$\mathcal{Z}_2^{32} = \{\zeta \in [0, 1] : I_H^{32}(\zeta) \subset (-c, c), I_L^{32}(\zeta) \subset (-c, c)\}.$$

Proposition 3. *Starting from the initial profile $(3, 2)$, the following statements hold.*

1. *The absorbing asymmetric region is ordered below either consensus region. If $\zeta_A \in \mathcal{Z}_A^{32}$ and $\zeta_H \in \mathcal{Z}_H^{32}$, then $\zeta_A < \zeta_H$. If $\zeta_A \in \mathcal{Z}_A^{32}$ and $\zeta_2 \in \mathcal{Z}_2^{32}$, then $\zeta_A < \zeta_2$.*
2. *The two consensus regions cannot both be nonempty under the same primitives: $\mathcal{Z}_H^{32} \neq \emptyset \implies \mathcal{Z}_2^{32} = \emptyset$, and $\mathcal{Z}_2^{32} \neq \emptyset \implies \mathcal{Z}_H^{32} = \emptyset$.*
3. *If $\zeta \in \mathcal{Z}_A^{32}$, then $(3, 2)$ is absorbing. Group H sends message 3 forever, and group L sends message 2 forever.*
4. *If $\zeta \in \mathcal{Z}_H^{32}$, then the path $(3, 2) \rightarrow (3, 3)$ is guaranteed pathwise. All agents send message 3 from date 1 onward.*
5. *If $\zeta \in \mathcal{Z}_2^{32}$, then the path $(3, 2) \rightarrow (2, 2)$ is guaranteed pathwise. All agents send message 2 from date 1 onward.*

Proof of Proposition 3

The block inputs from the initial profile $(3, 2)$ are $s_H^{32}(\zeta) = r(1 - \zeta(1 - q))$ and $s_L^{32}(\zeta) = \zeta qr$. The first function is strictly decreasing in ζ , and the second is strictly increasing in ζ .

First take $\zeta_A \in \mathcal{Z}_A^{32}$ and $\zeta_H \in \mathcal{Z}_H^{32}$. Since $\zeta_A \in \mathcal{Z}_A^{32}$, $s_L^{32}(\zeta_A) + \bar{\Delta} < c$. Since $\zeta_H \in \mathcal{Z}_H^{32}$, $(1 - \alpha)m_0 + \alpha(s_L^{32}(\zeta_H) - \bar{\Delta}) > c$. Because $m_0 < c$, this implies $s_L^{32}(\zeta_H) - \bar{\Delta} > c$. Hence $s_L^{32}(\zeta_A) < c - \bar{\Delta} < c + \bar{\Delta} < s_L^{32}(\zeta_H)$. Since s_L^{32} is strictly increasing, $\zeta_A < \zeta_H$.

Next take $\zeta_A \in \mathcal{Z}_A^{32}$ and $\zeta_2 \in \mathcal{Z}_2^{32}$. Since $\zeta_A \in \mathcal{Z}_A^{32}$, $s_H^{32}(\zeta_A) - \bar{\Delta} > c$. Since $\zeta_2 \in \mathcal{Z}_2^{32}$, $(1 - \alpha)h_0 + \alpha(s_H^{32}(\zeta_2) + \bar{\Delta}) < c$. Because $h_0 > c$, this implies $s_H^{32}(\zeta_2) + \bar{\Delta} < c$. Hence $s_H^{32}(\zeta_A) > c + \bar{\Delta} > c - \bar{\Delta} > s_H^{32}(\zeta_2)$. Since s_H^{32} is strictly decreasing, $\zeta_A < \zeta_2$.

We now show that $\mathcal{Z}_H^{32}$ and $\mathcal{Z}_2^{32}$ cannot both be nonempty. Suppose, toward a contradiction, that $\zeta_H \in \mathcal{Z}_H^{32}$ and $\zeta_2 \in \mathcal{Z}_2^{32}$. The H -block conditions imply $(1 - \alpha)h_0 + \alpha(s_H^{32}(\zeta_H) - \bar{\Delta}) > c$ and $(1 - \alpha)h_0 + \alpha(s_H^{32}(\zeta_2) + \bar{\Delta}) < c$. Hence $s_H^{32}(\zeta_H) > s_H^{32}(\zeta_2)$. Since s_H^{32} is strictly decreasing, $\zeta_H < \zeta_2$. The L -block conditions imply $(1 - \alpha)m_0 + \alpha(s_L^{32}(\zeta_H) - \bar{\Delta}) > c$ and $(1 - \alpha)m_0 + \alpha(s_L^{32}(\zeta_2) + \bar{\Delta}) < c$. Hence $s_L^{32}(\zeta_H) > s_L^{32}(\zeta_2)$. Since s_L^{32} is strictly increasing, $\zeta_H > \zeta_2$, a contradiction. Therefore the two consensus regions cannot both be nonempty.

It remains to prove the pathwise conclusions. If $\zeta \in \mathcal{Z}_A^{32}$, then under the profile $(3, 2)$, every H -agent's current input lies above c , while every L -agent's current input lies in $(-c, c)$. Since H -agents start in B_3 and L -agents start in B_2 , convexity of the update implies that all H -agents remain in B_3 and all L -agents remain in B_2 . The same argument applies at every later date, so $(3, 2)$ is absorbing.

If $\zeta \in \mathcal{Z}_H^{32}$, then both date-one score intervals are contained in B_3 . Hence all agents send message 3 from date 1 onward. Since all agents then receive social input r and $r - \bar{\Delta} > c$, high-message consensus is absorbing. If $\zeta \in \mathcal{Z}_2^{32}$, then both date-one score intervals are contained in B_2 . Hence all agents send message 2 from date 1 onward. Since all agents then receive social input 0 and $\bar{\Delta} < c$, interior-message consensus is absorbing. ■

The proposition shows that the main selection force does not depend on the lagging group starting from the lowest message. Starting from $(3, 2)$, group H already sends the state-aligned message, while group L has not yet crossed into that message. Raising ζ increases the lagging group's exposure to the state-aligned message but reduces the state-aligned group's exposure to its own message. Depending on which margin binds first, the society reaches either high consensus or interior consensus.

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